polarSA

Radiation pattern report

2026-10-29 06:53

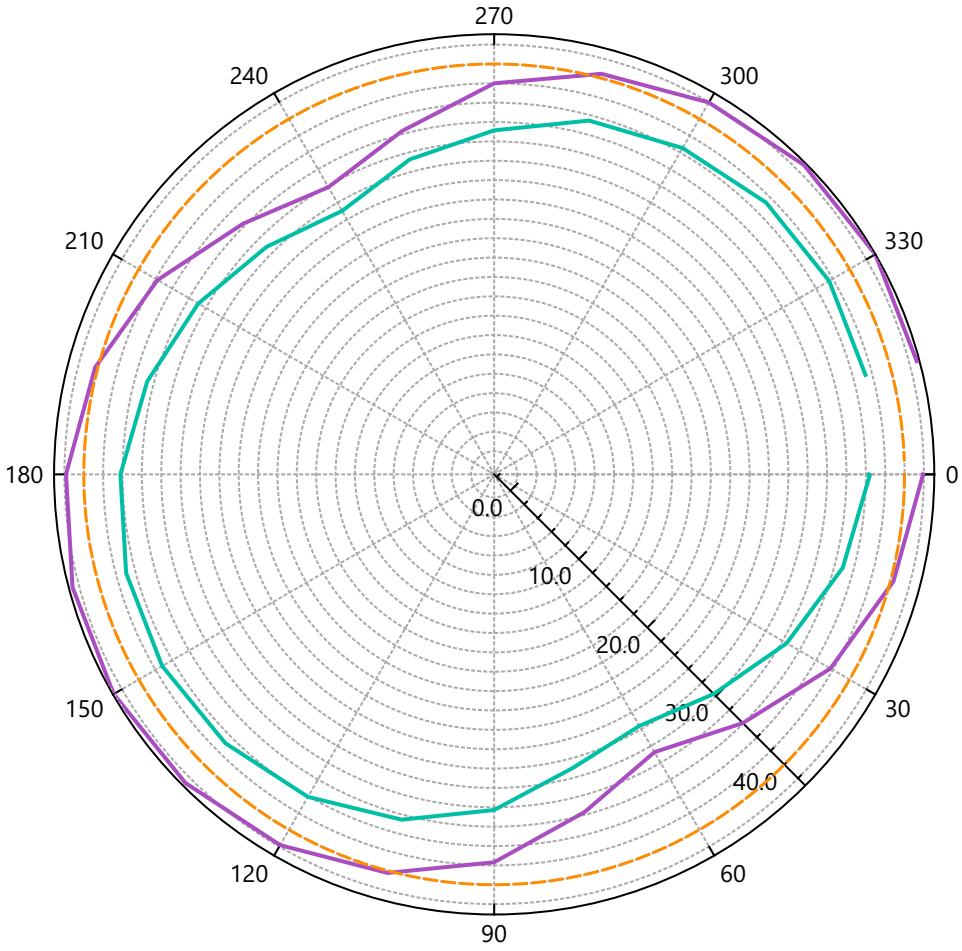
Config: session_emc_30-1000MHz.json

Frequency	144.000 MHz	**Scale**	absolute (dBµV/m)
Config file	session_emc_30-1000MHz.json	**Traces**	2
Levels	tinySA level (dBm) + calibration calibration_example_3m.csv (EXAMPLE values (fictitious broadband antenna + 3 m cable): 107 + AF + cable loss. Replace with your own setup calibration.) = dBµV/m	**Limits**	EN 55032 / CISPR 32 Class B radiated, 3 m (EN55032_ClassB_3m.csv, dBµV/m)
DUT	Sensor board rev B	**Antenna**	Broadband antenna, 3 m, horizontal
Polarization	Horizontal	**Operator**	Quantor Innovation

Pre-compliance scan, EN 55032 Class B at 3 m. 144 MHz re-measured in quasi-peak.

Live capture · Date: 2026-10-28T14:40:00 Center (MHz): 515.0000 Span (MHz): 970.0000 RBW: 850kHz Attenuation (dB): 0 LNA: on Ext gain dB: 0.0 Detector: peak Data unit: dBm (tinySA input level) Start angle (deg): 0 End angle (deg): 360 Angle step (deg): 15

Radiation pattern at 144.000 MHz



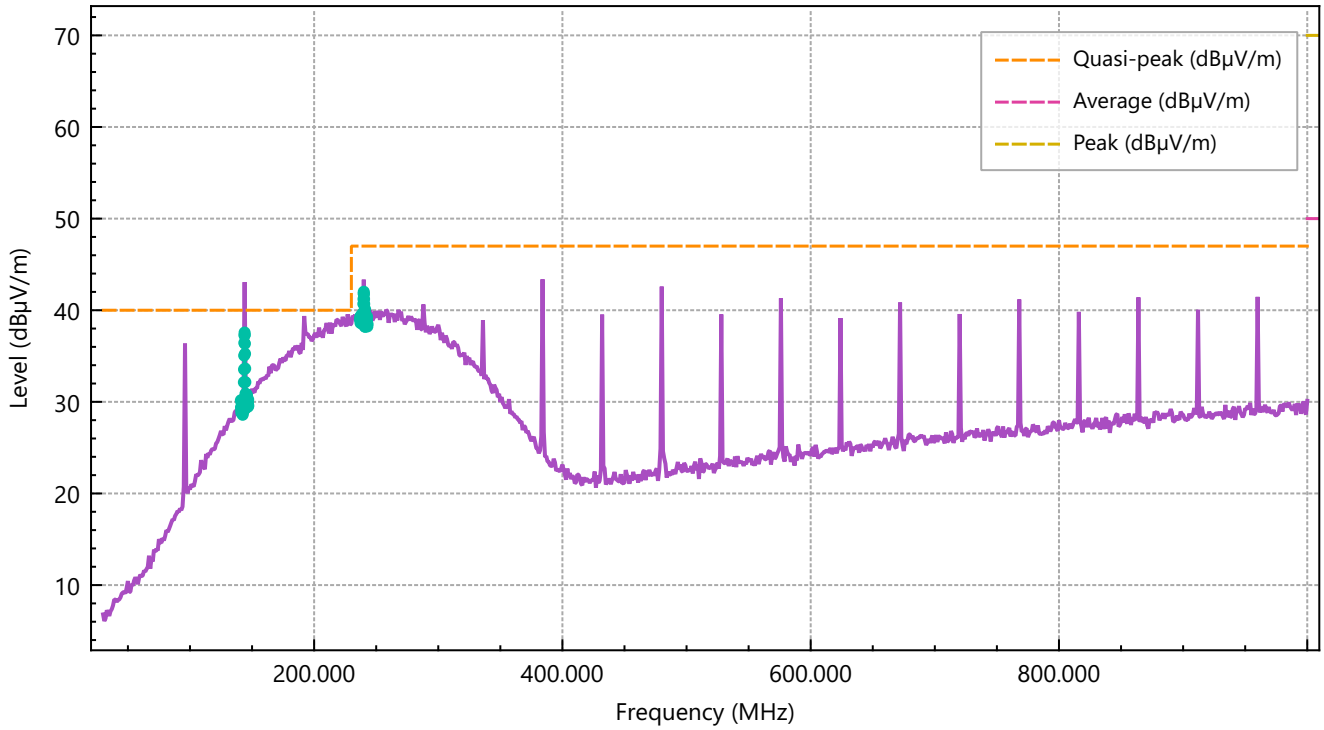
Trace	Max	Average	−3 dB BW	F/B	Rot. / gain
Live capture	43.1 dBµV/m @ 330°	40.3 dBµV/m	93.7°	0.1 dB	0° / +0.0 dB
emc_scan_quasi_peak.csv	37.6 dBµV/m @ 330°	34.9 dBµV/m	97.7°	0.4 dB	0° / +0.0 dB

Limit check (max hold over all angles)

Trace	Limit	Worst margin (negative = above the limit)
Live capture	Quasi-peak	-3.1 dB at 144.000 MHz (43.1 dBµV/m vs 40.0 dBµV/m)
Live capture	Average	+19.9 dB at 1000.000 MHz (30.1 dBµV/m vs 50.0 dBµV/m)
Live capture	Peak	+39.9 dB at 1000.000 MHz (30.1 dBµV/m vs 70.0 dBµV/m)
emc_scan_quasi_peak.csv	Quasi-peak	+2.4 dB at 144.000 MHz (37.6 dBµV/m vs 40.0 dBµV/m)

emc_scan_quasi_peak.csv	Average	no data in the limit's frequency range
emc_scan_quasi_peak.csv	Peak	no data in the limit's frequency range

Max emissions (max hold over all angles)



Trace details

Live capture

Live capture, not saved to a file

Date: 2026-10-28T14:40:00
Center (MHz): 515.0000
Span (MHz): 970.0000
RBW: 850kHz
Attenuation (dB): 0
LNA: on
Ext gain dB: 0.0
Detector: peak
Data unit: dBm (tinySA input level)
Start angle (deg): 0
End angle (deg): 360
Angle step (deg): 15

emc_scan_quasi_peak.csv

C:\Users\TECH1\Desktop\Local-max\max\PolarSA\PolarSA\examples\emc_scan_quasi_peak.csv

DUT: Sensor board rev B
Antenna: Broadband antenna, 3 m, horizontal
Polarization: Horizontal
Detector: quasi-peak (tinySA trace calc quasi: instant charge, slow discharge over successive sweeps; restarted from a raw sweep at each angle)
Comment: Synthetic example data generated for the polarSA documentation
Operator: Quantor Innovation
Date: 2026-10-28T15:05:00
RBW: 100kHz
Attenuation (dB): 0
LNA: on
Ext gain dB: 0.0
Data unit: dBm (tinySA input level)
Start angle (deg): 0
End angle (deg): 360
Angle step (deg): 15
Qp window: 141.0000 - 147.0000 MHz, 51 points, 20 sweeps per angle, RBW 100kHz
Qp window: 237.0000 - 243.0000 MHz, 51 points, 20 sweeps per angle, RBW 100kHz